

PORTABLE SERIES

Digital Balance - Operators Manual

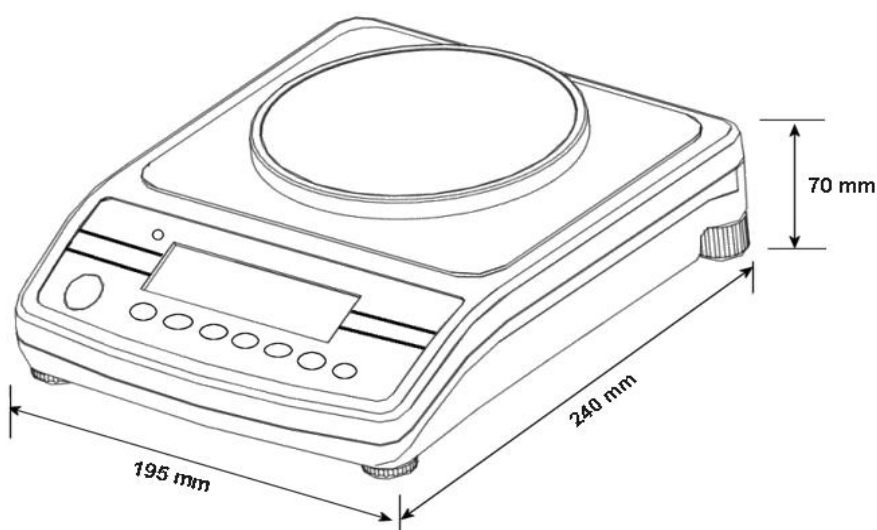
Rev 3.0 D June 2008

CONTENTS

INTRODUCTION.....	3
SPECIFICATIONS	4
TECHNICAL SPECIFICATIONS	4
COMMON SPECIFICATIONS.....	4
INSTALLATION.....	5
LOCATING THE BALANCE	5
SETTING UP THE BALANCE.....	5
BATTERY CHARGING.....	6
DISPLAY	6
KEY DESCRIPTIONS	7
OPERATION	8
ZEROING THE DISPLAY	8
TARING.....	8
WEIGHING.....	9
PERCENT WEIGHING.....	9
PARTS COUNTING	10
ACCUMULATION.....	10
PARAMETERS	11
SELECTING THE UNITS	12
SETTING OF THE BACKLIGHT	13
SETTING THE PRINTING PARAMETERS	14
Selecting the communication port	14
Setting the print mode	14
Setting the baud rate	14
Parity settings.....	15
SETTING OF AUTO-SWITCH-OFF.....	15
TECHNICAL PARAMETERS	15
CALIBRATION	16
COMMUNICATION WITH A COMPUTER/PRINTER.....	16
ERROR MESSAGES	17

INTRODUCTION

- The portable series of digital balances are simple and easy to use.
- The balances can be operated with an A.C. adapter or using the internal rechargeable battery.
- The weighing process is fast and accurate, making it suitable for the general purpose weighing applications, parts counting, percentage weighing functions, etc.
- There are 5 models in this series with capacities ranging from 120 grams to 4000 grams.
- All models include automatic zero tracking, automatic tare, accumulation facility that allows the weight to be stored and recalled as an accumulated total and bi-directional RS-232 interface for communicating with a PC or printer.
- All have stainless steel weighing platforms on an ABS plastic base assembly.
- Keypads are sealed with colour coded membrane switches.
- The balances have an optional weigh below facility.
- The displays are large, easy to read liquid crystal type displays (LCD). It is provided with a backlight and also a bar-graph at the bottom to display the weight of the item on the pan with respect to the capacity of the balance.
- The LCD normally shows the weight and the units of the weight selected previously (Grams, Carats, Pounds, etc). See section 5.1 for details on Units.



SPECIFICATIONS

TECHNICAL SPECIFICATIONS

	120g	400g	800g	2000g	4000g
Maximum Capacity	120g	400g	800g	2000g	4000g
Readability	0.005g	0.01g	0.01g	0.05g	0.1g
Tare Range	120g	400g	800g	2000g	4000g
Repeatability(s.d.)	0.005g	0.01g	0.01g	0.05g	0.1g
Linearity ±	0.01g	0.02g	0.02g	0.1g	0.2g
Pan	Φ120mm (4.7")				124×144mm (4.9"X5.7")
Draft shield	Included as standard				
Units of Measure	g / ct / Lb / OZ / d / GN / OZt / dWt / MM / TL.T / TL.C / TL.t / t / N /g2.				

COMMON SPECIFICATIONS

Interface	RS-232, bi-directional
Stabilisation Time	2 Seconds typical
Operating Temperature	0°C - 40°C / 32°F - 104°F
Power supply (external)	9VDC / 800 mA
Calibration	Automatic External
Display	16mm high 6 digits LCD With auto backlight and loading bar graph
Balance Housing	ABS Plastic with Stainless Steel platform
Overall Dimensions (wxdxh)	195 x 240 x 70mm (7.7" x 9.5" X 2.8")
Net Weight	1.5 kg (3.3 lb.)

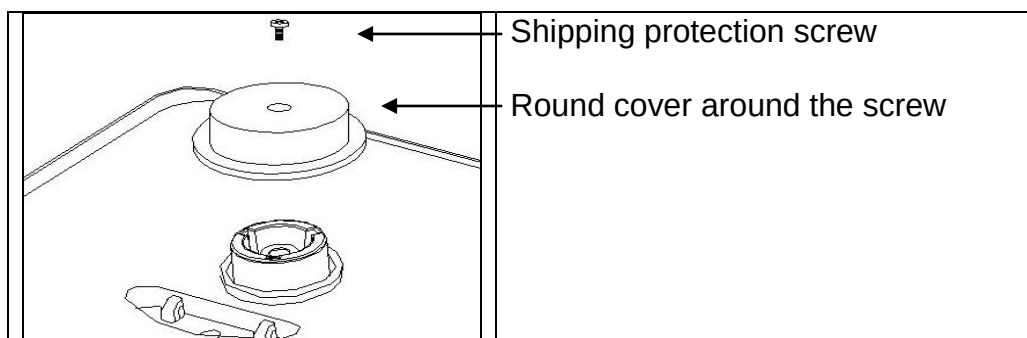
INSTALLATION

LOCATING THE BALANCE

- Avoid extreme temperatures. Do not place the balance in direct sunlight or near air conditioning vents. Avoid air movement such as from fans or opening of doors and windows.
- Avoid unsuitable tables. The table or floor must be rigid and not vibrate. Do not place near vibrating machinery.
- Avoid unstable power sources. Do not use the balance near any large user of electricity such as welding equipment or large motors.
- Avoid high humidity that might cause condensation. Avoid direct contact with water. Do not spray or immerse the balances in water.
- Keep the weighing area clean.
- Do not stack material on the balance when not in use.
- Protect the balance from extreme temperatures, vibration and dust.

SETTING UP THE BALANCE

- Remove the shipping protection screw and round cover as shown below using the Allen key supplied in the box. (It is advisable to keep the shipping protection screw and the round cover along with the packing box for using during shipping to another location in future).

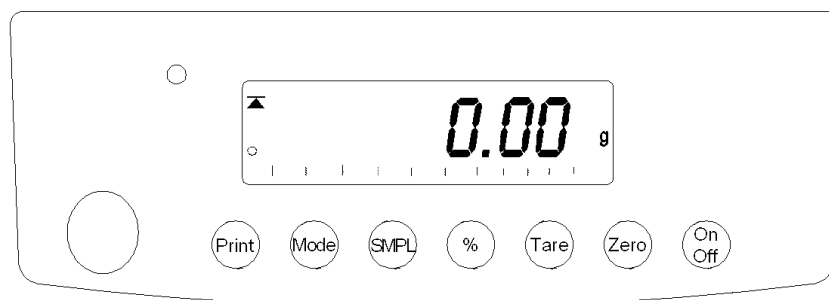


- The balance comes with a stainless steel platform, packed separately.
- Place the platform in the receptacles on the top cover.
- Do not press with excessive force as it could damage the load cell inside.
- Level the balance by adjusting the four feet. The balance should be adjusted such that the bubble in the spirit level is in the centre of the level and the balance is supported by all four feet.
- Turn the power on by pressing the **[On/Off]** switch.
- The unit will display the model number and the software revision number for a while and then will count down performing a self-test.
- When ready the display will show zero along with the weighing unit selected during the last operation.

BATTERY CHARGING

- The balances can be operated from the internal rechargeable battery or using an A.C. Adapter. The battery life is approximately 24 hours depending on the usage of the backlight.
- The display will show an indication when the battery needs to be charged. To charge the battery, plug in the AC adapter at the back of the balance and attach it to the main power supply. The charging indicator at the left top corner of the LCD will be on.

DISPLAY



KEY DESCRIPTIONS

KEYS	PRIMARY FUNCTION	SECONDARY FUNCTION
[On/Off]	To turn the balance On or Off.	----
[Zero]	Sets the zero point for all subsequent weighing. The display shows zero.	A secondary function is of " Enter " key for setting parameters or other functions.
[Tare]	Tares the balance. Stores the current weight in memory as a tare value, subtracts the tare value from the weight and shows the results. This is the net weight.	Pressing [Tare] again will cancel the tare value.
[%]	Enters the percent weighing function.	When in percentage mode, press [%] key to return to normal weighing mode.
[Smpl]	Use this key to enter counting mode from the normal weighing mode,	A secondary function allows the user to return to normal weighing mode while in counting mode. It also clears the accumulated value from the memory when pressed during recalling the total.
[Mode]	Pressing this key will cycle through the weighing units which are enabled. It also scrolls through the options during setting up a parameter.	Displays the unit weight , total weight and the count when in the parts counting mode.
[Print]	To print the results to a PC or printer using RS-232 (A USB interface will be available shortly). It also adds the value to the accumulated memory if the accumulation function is not automatic.	Secondary function is to return to normal operation when the balance is in a parameter setting mode.

OPERATION

ZEROING THE DISPLAY

- Press the **[Zero]** key when the platform is empty.
- **[Zero]** key may be pressed at any time to set the zero point from which all other weighing and counting is measured.
- When the zero point is obtained the “**O**” indicator will be on in the left bottom corner of the LCD.
- The balance has an automatic re-zeroing function to account for minor drifting or accumulation of material on the platform. However you may need to press the **[Zero]** key to re-zero the balance if small amounts of weight are shown when the platform is empty.

TARING

- Zero the balance by pressing the **[Zero]** key, if necessary.
- The “**O**” indicator will be on.
- Place a container on the platform, a value for its weight will be displayed.
- Press the **[Tare]** key to tare the balance. The weight that was displayed is stored as the tare value. It is subtracted from the displayed value, leaving zero on the display. The **NET** symbol will be on at the right top corner to indicate that the weight displayed is the net weight.
- As the product is added, only the net weight will be shown.
- The balance could be tared a second time if another type of product was to be added to the first one. Again only the weight that is added after taring will be displayed.
- When the container is removed a negative value will be shown. If the balance was tared just before removing the container, this value is the gross weight of the container plus all products those were removed.
- The “**O**” indicator will also be on because the platform is back to the same condition as it was when the **[Zero]** key was last pressed.
- To cancel the tare value, press **[Tare]** again.

WEIGHING

- After zeroing the display and taring the container, place the sample in the container to determine the weight of the sample. The weight will also be displayed through a bar-graph at the bottom of the LCD.
- The display will show the weight and the units of weight currently in use. The stable indicator  will be on. To change the weighing units, press **[Mode]** to cycle through the options of units those are enabled. See the Parameter section to enable or disable the weighing units.
- The accumulated value in the memory will be cleared when a new weighing unit or during recalling the memory, % weighing or parts counting is selected.

PERCENT WEIGHING

- The balance will allow a sample weight to be shown as 100%. Then any other weight placed on the balance will be displayed as a percentage of the original sample.
- For example is 350g is placed on the balance and the **[%]** key is pressed the display will show 100.00%. Removing the 350g weight and place a 300g weight on the balance. The display will show 85.71% as 300g is 85.71% of 350g.
- **Note:** The balance may jump by large numbers unexpectedly if small weights are used to set the 100% level. For example, if only 23.5g is on a balance with 0.5g increments and the balance is set to 100%, the display will show 100.00%. However, a small change of weight will cause the display to jump to 102.13%, as one division (0.5g) increase to 24.0g will be equivalent to a 2.13% increase.
- Pressing the **[%]** key again will return the balance to normal weighing.

PARTS COUNTING

- When the balance is in normal weighing mode, pressing the **[Smpl]** key will start the parts counting function.
- Before beginning, tare the weight of any container that will be used, leaving the empty container on the balance.
- Place the number of samples on the balance. The number should match the options for parts counting, 10, 20, 50, 100 or 200 pieces.
- Press the **[Smpl]** key.
- The balance will initially show "**SP 10**" asking for a sample size of 10 parts. Press **[Mode]** to cycle through the options: 10, 20, 50, 100, 200 and back to 10.
- Press **[Smpl]** again when the number matches the number of parts used for the sample. As more weight is added the display will show the number of parts (**PCS**) with respect to the chosen sample size.
- Press **[Mode]** to display unit weight (**W/P**), total weight (g) or the count (**PCS**). The total and the unit weight are shown in the current units.
- Press **[Smpl]** to return to normal weighing.

ACCUMULATION

- When the balance is set for automatic accumulation, the weight will be added to the memory when the balance becomes stable. **The balance must return to zero before another weight is added.**
- When the balance is set to manual accumulation (See the Parameter Section) the weight displayed will be stored in memory when the **[Print]** key is pressed and the weight is stable.
- The display will show "**ACC 1**" followed by the total value in the memory for 2 seconds before returning to displaying the weight of the item on the balance.

- The weight will be output to a printer or PC.
- Remove the weight, allowing the balance to return to zero and put a second weight on.
- Press **[Print]**, the display will show "**ACC 2**" followed by the new total in the memory for 2 seconds before returning to displaying the weight of the item on the balance.
- Continue until all weights have been added.
- To view the totals in memory press the **[Print]** key when the balance is at zero. The display will show "**ACC xx**" (where "**xx**" is the total number of readings) and the total weight, before returning to zero.
- The total will also be printed via the RS-232 interface.
- To clear the memory, press **[Smpl]** immediately after pressing **[Print]**.

PARAMETERS

- The balance has 5 parameters that can be chosen by the user to set the balance.
- To set the parameters press the **[Mode]** key during self-checking test.
- After a few seconds, the display will show the first function "**F1 UNT**".
- To view the current settings of each parameter, press the **[Zero]** key.
- Press the **[Zero]** key again to cycle through the other parameters along with their current settings. For example, if "**OFF**" is displayed with the weighing unit Pounds, the user will not be able to use this unit while weighing a sample.
- Press **[Mode]** to change the settings of a particular parameter. For example, to enable the weighing unit pounds change the setting to "**on**" by pressing **[Mode]** now.
- Use the **[Print]** key to leave a parameter unchanged.

The following parameters are available:

FUNCTION	DESCRIPTION
F1 UNT	Sets the units to be used (See section 5.1) g / ct / Lb / OZ / d / GN / OZt / dWt / MM / TL.T / TL.C / TL.t / t / N / g2.
F2 EL	Sets the backlight (See section 5.2) EL on : backlight always on EL AU : backlight automatically turns on when a key is pressed EL off : backlight always off
F3 SEr	Sets the print parameters (See section 5.3)
F4 off	Sets the auto switch-off parameter (See section 5.4)
TECH	Pressing [Zero] will allow the user to enter the technical parameters setting mode. It is recommended not to proceed with the settings of these parameters as those are set up at the factory.

SELECTING THE UNITS

- When “**F1 Unit**” is displayed, press the **[Zero]** key to view the current settings of each unit.
- Press **[Zero]** again to cycle through the other units along with their current settings. For example, if “**off**” is displayed with the weighing unit Carats, the user will not be able to use this unit while weighing.
- Press **[Mode]** to change the settings of a particular unit. For example, to enable the weighing unit Carats, change the setting to “**on**” by pressing **[Mode]** now.
- Use the **[Print]** key to leave a parameter unchanged.
- While weighing a sample, the display will show the weight and the units of weight currently in use. To view the weight in other units (which are enabled), press **[Mode]**.
- The following table shows different units which are available to the user and the conversion factors for each.

Sl. No.	Name of the Units	Description	Conversion Factor	Display Symbol
01	Grams	A standard metric unit	1.0	g
02	Carats	Used for weighing jewellery and gems, etc.	5.0	ct
03	Pounds	Standard weighing unit in UK/USA.	0.002205	Lb
04	Ounce	Avoirdupois ounce. 16 ounces make a pound.	0.03528	OZ
05	Drams	An ancient unit of weight. Equals to 1/16 th of an ounce.	0.5645	d
06	Grains	A basic weighing unit in the imperial system. Used to weigh gun powder.	15.432	GN
07	Ounce Troy	Troy ounce- used for weighing gold, silver and in pharmacy.	0.03216	OZt
08	Penny-weight	Pennyweight was the weight of a silver penny in medieval England. Equals to 1/20 th of an Ounce Troy.	0.6432	dWt
09	Mommes	A weighing unit used in Japan to weigh pearls.	0.26667	MM
10	Taels Hk.	Hongkong Taels- used to weigh coral, pearls, etc.	0.02675	TL.T
11	Taels S.	Singapore Taels	0.02646	TL.C
12	Taels T.	Taiwan Taels	0.02675	TL.t
13	Tical	An Asian weighing unit	0.08576	t
14	Newtons	Used to measure force	0.009808	N
15	Grams	Grams with last digit suppressed	1.0	g2

SETTING OF THE BACKLIGHT

- The backlight may be enabled or disabled by the user.
- If the backlight is disabled, the battery life will be maximised.
- The following settings are available:

EL AU	Sets the backlight to operate automatically when a weight is placed on the balance or a key is pressed.
EL Off	Sets the backlight to be off.
EL On	Sets the backlight to be on for full time.

- When “**F2 EL**” is displayed, press the [**Zero**] key to view the current setting for the backlight.
- Press [**Mode**] to scroll through other settings.
- Press [**Zero**] to store a particular setting.
- Use the [**Print**] key to leave a parameter unchanged.

SETTING THE PRINTING PARAMETERS

When “**F3 SEr**” is displayed, press [**Zero**] to view the first printing parameter.

Selecting the communication port

- The selected port, for example “**s 232**” will be displayed.
- Press [**Mode**] to cycle through the other options. The user can select “**s 232**” or “**s USB**” as the communication port. Currently, only RS-232 interface is available and the balance is set with this port only.
- Press [**Zero**], to view the next printing parameter.

Setting the print mode

- The current setting of the print mode is displayed,
- Press [**Mode**] to cycle through the other options.
- The following options are available:

P1 Prt	The balance sends the accumulated weights to the communication port, whenever the [Print] key is pressed.
P2 Con	It will set the RS-232 interface to print the weights continuously and the accumulation function is disabled.
P3 AUT	The weighing results will be sent to the communication port automatically whenever a weight is placed on the platform.

- Press [**Zero**], to view the next printing parameter.

Setting the baud rate

- The current baud rate setting will be displayed.
- Press [**Mode**] to cycle through the other options.
- The following settings are available:

b 600
b 1200
b 2400
b 4800 (default value)
b 9600

- Press [**Zero**] to view the next printing parameter.

Parity settings

- The current Parity setting will be displayed.
- Press **[Mode]** to cycle through the other options.
- The following settings are available:

8 n 1	8 data bits, no parity
7 E 1	7 data bits, even parity
7 0 1	7 data bits, odd parity

- Press **[Zero]**, the display will go back showing "**F3 Ser**".
- Press **[Mode]** to go to the next parameter which is "**F4 off**".

SETTING OF AUTO-SWITCH-OFF

- The Auto switch-off time may be set up by the user.
- When "**F4 off**" is displayed, press the **[Zero]** key to view the current setting for the backlight.
- Press **[Mode]** to scroll through other settings (0, 5, 10, 20 and 30 minutes).
- Press **[Zero]** to store a particular setting.
- Use the **[Print]** key to leave a parameter unchanged.

TECHNICAL PARAMETERS

- Next parameter is "**TECH**". This allows the user to enter the Technical Parameters section.

Note: It is recommended not to proceed with the settings of the Technical Parameters which are set up at the factory and are often referred to as Service Parameters. Check the Service Manuals for more details.

- Press **[Print]** to return to weighing. The balance will run the self-test before displaying the zero along with the last selected unit.

CALIBRATION

- This parameter allows the user to calibrate the balance any time during the operation.
- Switch the balance off and then on it again by pressing **[On/Off]**. During the self-test press **[Smpl]** and **[Print]** together.
- Display will show **"unLoAd"**. Remove all weights from the pan.
- Wait for the stability indicator to be on and then press **[Zero]**.
- The display will then show the first calibration weight request.
- Press **[Mode]** to scroll through the other calibration weights.
- Press **[Zero]** to select calibration weight. The display will show **"LoAd"**.
- Place this weight on the platform and press **[Zero]** when the stability indicator will be on to show the value is stable.
- If the calibration is complete, the balance will show **"PASS"** and will return to normal weighing counting back to zero. Remove the calibration weight at this stage.
- If the wrong calibration weight is used, the balance will show **"FAIL H"** (when the weight is higher) or **"FAIL L"** (when the weight is lower). Repeat the process using the correct calibration weight.

Calibration weights:

Model #	120g	400g	800g	2000g	4000g
Weight 1	40g	100g	200g	500g	1000g
Weight 2	80g	200g	500g	1000g	2000g
Weight 3	120g	400g	800g	2000g	4000g

COMMUNICATION WITH A COMPUTER/PRINTER

The Portable Digital Series of Balances come with a standard RS-232 interface.

The standard Interface parameters are: Connection details are:

RS-232 output of weighing data
ASCII code
4800 Baud
8 data bits
No Parity

Connector: 9 pin d-subminiature socket
Pin 3 Output
Pin 2 Input
Pin 5 Signal Ground

Data Format for normal weighing operations, parts counting or recalling of totals from memory will all be different.

Normal Output: G S _ X X X . X X u u u

GS	123.45g	GS for Gross Weight, NT for Net Weight and u for unit of weight
No.	1	This number increments every time a new value is stored in memory
Total	123.45g	The total value stored in memory
<lf>		Includes 2 line feeds
<lf>		

Input commands format:

The balance can be controlled with the following commands. The commands must be sent in upper case letters, i.e. “T” not “t”.

T<cr><lf>	Tares the balance to display the net weight. This is the same as pressing [Tare] key.
Z<cr><lf>	Sets the zero point for all subsequent weighing. Display shows zero.
T5.345<cr><lf>	Would be same as entering a preset tare value of 5.345 from keypad
P<cr><lf>	Prints the results to a PC or printer using the optional RS-232 interface. It also adds the value to the accumulation memory if the accumulation function is not set to automatic.

ERROR MESSAGES

During the initial power-on testing or during operation it is possible that the scale may show an error message. The meaning of the error messages is described below.

ERROR CODE	DESCRIPTION	POSSIBLE CAUSES
Err 4	Initial Zero is greater than allowed (4% of maximum capacity) when power is turned on or when the [Zero] key is pressed,	- Weight on the pan when turning on. - Excessive weight on the pan when zeroing the balance. - Improper calibration of the balance. - Damaged load cell. - Damaged Electronics.
Err 5	Keyboard Error.	Improper operation of the balance.
Err 6	A/D count is not correct when turning the balance on	- Load cell damaged. - Electronics damaged.
Err 9	A/D count is not stable when turning the balance on	- Initial instability during turning on which will get adjusted automatically - There may be movement, vibration or dirt on balance during turning on.

If an error message is shown, repeat the procedure that caused the message such as turning the balance on, calibration or any other functions. If the error message is still shown, contact your dealer for further support.



Manufacturer's Declaration of Conformity

This product has been manufactured in accordance with the harmonised European standards, following the provisions of the below stated directives:

Electro Magnetic Compatibility Directive 89/336/EEC

Low Voltage Directive 73/23/EEC

FCC COMPLIANCE

This equipment has been tested and found to comply with the limits for a Class A digital device, pursuant to Part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment. The equipment generates, uses, and can radiate radio frequency energy and, if not installed and used in accordance with the instruction manual, may cause harmful interference to radio communications. Operation of this equipment in a residential area is likely to cause harmful interference in which case the user will be required to correct the interference at his own expense.

Shielded interconnect cables must be employed with this equipment to insure compliance with the pertinent RF emission limits governing this device.

Changes or modifications not expressly approved by the manufacturer could void the user's authority to operate the equipment.

The digital balances have a 2 year warranty against manufacturing defects during normal usage as determined by the manufacturer and do not cover misuse or accident, acts of god, foreign matter or material getting inside the balance, powers surges, exposure to radioactive or corrosive materials, or any event not deemed to be as a result of the manufacturing process.

© Copyright by the manufacturer of these scales. All rights reserved. No part of this publication may be reprinted or translated in any form or by any means without the prior permission of the original manufacturer.

The manufacturer reserves the right to make changes to the technology, features, specifications and design of the equipment without notice.

All the information contained within this publication was to the best of our knowledge timely, complete, and accurate when issued. However we are not responsible for misimpressions which may result from the reading of this material.

